04/23

Wednesday, 23 April 2025 9:33 AM

→ Velocity BC on wall

→ Velocity BC on inviscid wall

→ Pressure & T

Inflow :

Incompressible flow → harder to model since no time derivative
(speed of sound is infinite → any change is
affected everywher

Last class

Sunday, April 27, 2025 2:37 PM

Boundary conditions

↳ Dirichlet boundary conditions

Prime factorization

↓

Problem 1 from HW7

Flux vector splitting:

↳ spectral radius

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

$$c = c^+ + c^-$$

Local eigenvalue

$$c^+ = \frac{1}{2}(c + |c|)$$

$$c^- = \frac{1}{2}(c - |c|)$$

↓

less dissipating

→ First order upwind →

Central differencing → fourth order diffusion (& coefficient has to be negative for stability)

Given PDE:

1) Δu^{n+1} everywhere Δ onset

2) Linearize

$$D_x \cdot (u^{n+1})^2 = (u^n)^2 + 2u^n \Delta u^{n+1}$$

$$\underbrace{D_x D_y (u^{n+1}(x,y))}$$

$$u^{n+1} = u^n + \Delta u^{n+1} \approx u^n + \Delta u^n$$

$$\frac{\partial}{\partial x}\left(\alpha \frac{\partial \phi}{\partial x}\right) = \frac{\left(a \frac{\partial \phi}{\partial x}\right)_{i+\frac{1}{2}} - \left(a \frac{\partial \phi}{\partial x}\right)_{i-\frac{1}{2}}}{\Delta x}$$

$$= \frac{a_{i+\frac{1}{2}}\left(\frac{\phi_{i+1} - \phi_i}{\Delta x}\right) - a_{i-\frac{1}{2}}\left(\frac{\phi_i - \phi_{i-1}}{\Delta x}\right)}{\Delta x}$$

Boundary conditions for tridiagonal matrix

$$L_{ij} \Delta u^*_{i+1,j} + M_{i,j} \Delta u^*_{i,j}$$





→ Dirichlet → does not change w/ time

$T = 40\,C$

$\Delta u^{n}_{i+1,j} = 0$ (does not change)

→ Neumann → $\frac{\partial u}{\partial x} = q$

$\frac{u_2 - u_1}{\Delta x} = q \rightarrow u_1 = u_2 - q\Delta x$

$M \rightarrow \frac{V}{C}$

$T \rightarrow C = \sqrt{\gamma R T}$

V

T

$\frac{\partial^2 P}{\partial x^2} = 0$

$$\frac{P_2 - 2P_1 + P_0}{\Delta x^2} = 0 \rightarrow P_0 = 2P_1 - P_2$$